

Materials Science: Team Project

Subject: What is a “BCC–Superalloy”?

Motivation: Body-centred-cubic BCC-superalloys are a nascent materials class following a microstructure template of a disordered BCC matrix (e.g. refractory metal, Fe, Ti) reinforced by coherent precipitates that adopt a BCC–derived ordered superlattice structure, to enable enhanced high temperature performance. This is by analogy to the highly successful Ni–based superalloys whose properties are underpinned by their microstructure template of γ/γ' FCC–Ni solid solution and ordered Ni_3Al . BCC-superalloys represent a powerful Beyond Ni–based superalloy design approach for high–temperature applications from aerospace engines and rockets to fusion energy. Various BCC-superalloy base systems offer increased melting points, oxidation/corrosion/irradiation resistance, reduced density and/or cost advantages. However, challenges exist in phase stability, low temperature ductility, oxidation/corrosion resistance and the availability of predictive tools.

Task Description: The aim of this task is to acquire knowledge and evaluate material design concepts as well as mechanical Properties of BCC-superalloys. The work will involve:

- Literature review of the alloy concepts and design of BCC-superalloys.
- A summary of specific properties of BCC-superalloys covering modelling, phase transformations, lattice misfit, mechanical behavior, deformation mechanisms, oxidation/corrosion and manufacturing.
- Discussing these properties by introducing an overview of at least three alloy designs of the BCC-superalloys type of class.

Expected Outcome: By completing this task, students will gain:

- A deeper understanding of new materials concepts beyond Ni–based superalloys
- Gain a critical insight into challenges to design new structural materials for high–temperature applications.
- Insight into materials’ design concepts and microstructure–property relations of high–temperature materials.

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